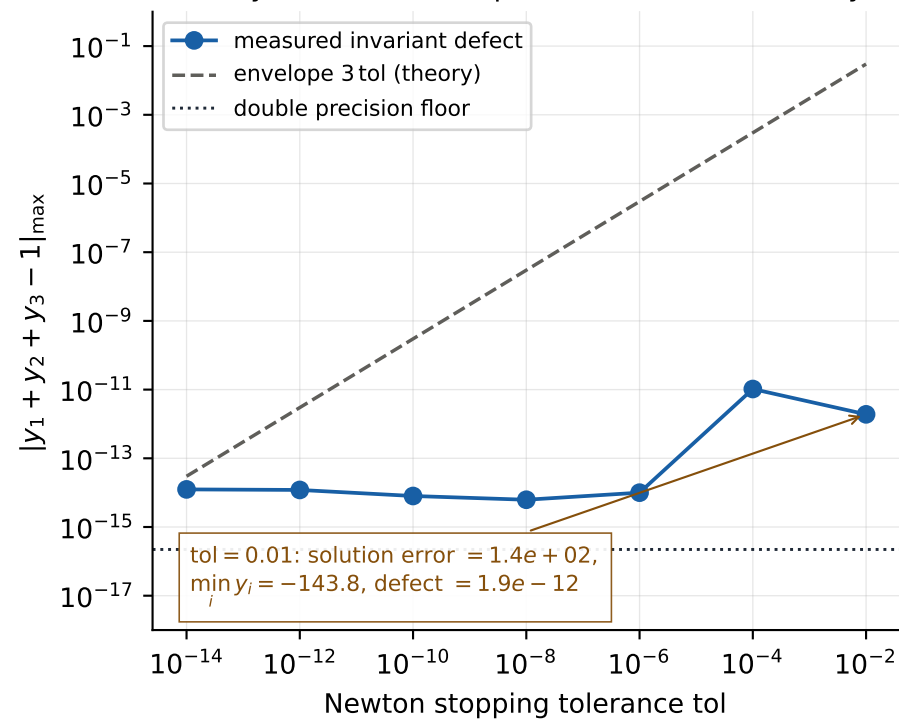
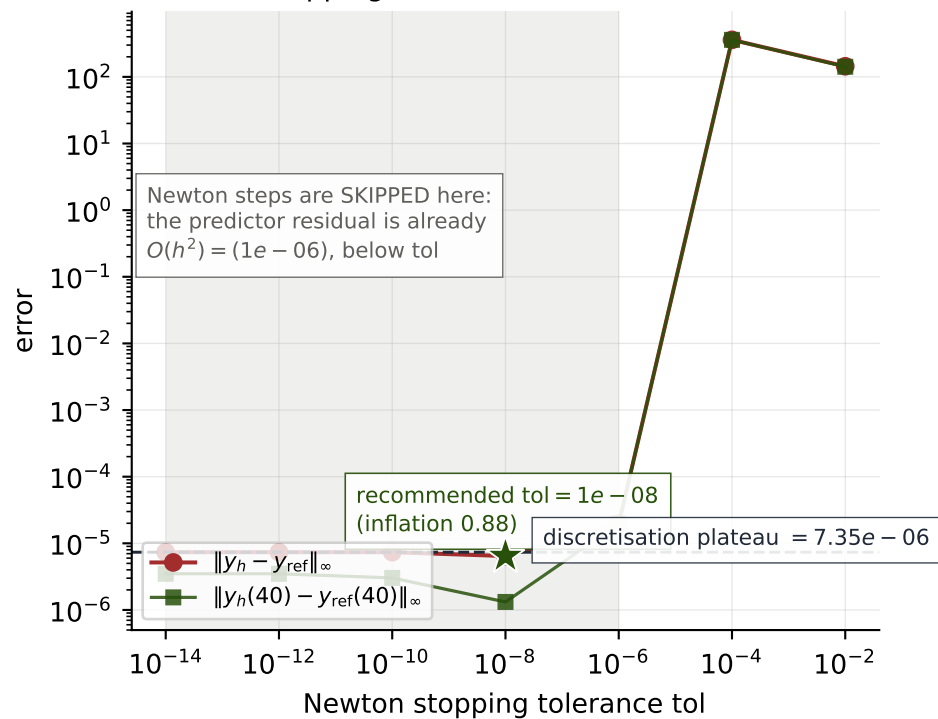


F10 Newton stopping-tolerance sweep at $h = 0.001$: the algebraic error must be negligible against the discretisation error

(a) The invariant defect cannot see the algebraic error
every Newton iterate preserves the sum exactly



(b) ... but the solution error does
the stopping rule must sit left of the break



(c) Newton is often FREE, and sometimes skipped
tight tolerances average about one update per step

